Student Performance
Early-Warning Prediction System

A Professional Applied Machine Learning Project in Educational Data Mining

Dataset:	20,000 students $\times$ 41 variables
Primary task:	Post-midterm Pass/Fail early-warning classification
Secondary task:	Final score regression
Deliverables:	Cleaned dataset, ML pipeline, Power BI dashboard

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Executive Summary

This report documents the end-to-end development of a **student early-warning prediction system**: a decision-support tool that identifies, at the post-midterm point in a semester, which students are at risk of failing their course. The project spans four stages — rigorous data cleaning and governance, exploratory data analysis with a formal data-leakage audit, comparative machine learning modeling, and deployment via an interactive Power BI dashboard.

Working with a dataset of 20,000 students and 41 variables, the project's central, somewhat counter-intuitive finding is that **most demographic and psychological variables carried negligible predictive signal**, while real signal concentrated in just four features: midterm score, quiz average, lecture attendance rate, and assignment submission rate. A second key finding was that a **simple, interpretable Logistic Regression model outperformed a tuned LightGBM gradient boosting model on every evaluation metric** for the primary classification task — evidence that model complexity should always be justified by the data, not assumed. The final recommended model achieves a recall of **88.1%** on the held-out test set, meaning it correctly identifies almost nine in ten at-risk students before the final exam.

1 Introduction

1.1 Problem Statement

Academic institutions routinely accumulate large volumes of student data — attendance, assessment scores, demographics, and behavioral indicators — but this data is rarely turned into timely, actionable insight for academic advisors. The central question this project addresses is:

> *"Can we reliably identify students at risk of failing a course early enough in the semester for meaningful intervention to still be possible?"*

The project is explicitly framed as a **decision-support tool**, not an automated determination of a student's outcome. Predictions are intended to route attention toward students who may benefit from advisor contact — the final decision and any intervention remain human-driven.

1.2 Project Scope

The work covers the complete applied machine learning lifecycle:

1. Data quality auditing and governed cleaning
2. Exploratory data analysis and statistical verification of assumed relationships
3. A formal data-leakage audit and prediction-time-cutoff definition
4. Feature engineering, grounded in tested (not assumed) justification
5. Comparative modeling: baseline, ensemble, and gradient-boosting classifiers
6. Cross-validation, hyperparameter optimization, and rigorous evaluation
7. Error analysis, explainability (SHAP), and subgroup fairness checks
8. Deployment as an interactive Power BI dashboard for non-technical stakeholders

1.3 Dataset Overview

The source dataset, `student_data_cleaned.xlsx`, contains **20,000 student records and 41 variables** spanning demographics, prior academic history, study behavior, attendance and engagement metrics, psychological/behavioral indicators, course characteristics, and final academic outcomes. No missing values or duplicate records were present in the source file at the point of ingestion into the modeling pipeline.

2 Data Cleaning & Governance

Before any modeling began, the dataset underwent a documented, decision-by-decision cleaning process. This section summarizes the governance framework applied.

2.1 Cleaning Methodology

The cleaning process followed a ten-step framework:

1. **Understand the dataset** — load, inspect shape, dtypes, and column meanings before any transformation.
2. **Check data quality** — scan for missing values, duplicate rows, wrong data types, inconsistent text, invalid values, and outliers.
3. **Clean text columns** — trim whitespace, standardize blank values, and convert placeholder strings (`"NA"`, `"N/A"`, `"null"`) into proper missing-value markers.
4. **Validate numerical columns** against realistic bounds (Table 1).
5. **Handle missing values** using a method justified per column — never a blanket zero-fill.
6. **Handle duplicates** — remove only genuine duplicate observations.
7. **Handle outliers** using IQR, Z-score, or domain rules, with the explicit principle that *an outlier is not automatically an error* and must be investigated before removal.
8. **Check cross-column consistency** — e.g. attendance cannot exceed 100%, scores cannot be negative.
9. **Recheck the cleaned dataset** — re-run every quality check post-cleaning.
10. **Export** the validated, clean file.

Table 1: Sample numerical validation rules applied during cleaning

Column	Valid Range
Age	Reasonable student-age range
Attendance Rate	0–100%
Exam Score	0–100
Stress Level	0–10
Sleep Hours	Reasonable range
GPA	Valid GPA scale
Course Difficulty	1–5

2.2 Missing Value Strategy

Missing-value handling was decided **per column**, not applied uniformly:

- **Numerical / ordinal columns** (age, parent income, number of siblings, family support, commute time, previous GPA, number of failed courses, high school grade, and the remaining numerical/ordinal fields): imputed with the **column median**, chosen specifically because the median is far less sensitive to extreme or unusual values than the mean — an important property for skewed real-world variables like income or commute time.
- **Categorical columns** (gender, part-time job status, course type): imputed as **"Unknown"** rather than an arbitrary mode value, to avoid silently fabricating a category membership that was never observed.

- **Outcome / target variables** (final_score, final_grade, pass_fail): **deliberately not imputed under any circumstance.** These are the variables the entire analysis exists to understand — inventing their values would silently corrupt every downstream conclusion. Records with missing outcomes were instead flagged and excluded only from analyses that specifically require the outcome, preserving them for outcome-independent analyses.

2.3 Worked Example: Age Column

As a representative case, the age column was cleaned as follows:

1. Missing values in age were identified.
2. A valid age range for students was defined.
3. Values outside that reasonable range were treated as missing (rather than kept as-is), since an impossible age is equally as unusable as a blank one.
4. The median age was computed from the remaining valid records.
5. Missing/invalid values were replaced with that median.
6. Post-cleaning, zero missing values remained in age.

This same audit-then-decide pattern — identify, validate range, choose a justified imputation method, verify zero remaining issues — was applied consistently across all 38 non-outcome columns.

3 Dataset Audit

3.1 Structural Summary

Table 2: Dataset structural summary

Property	Value
Rows	20,000
Columns	41
Missing values	0
Duplicate rows	0
Numeric columns	36
Categorical columns	5 (gender, part_time_job, course_type, final_grade, pass_fail)
Identifier column	None present

3.2 Target Variable Distributions

Three candidate target variables were profiled before selecting the modeling task.

Table 3: final_score descriptive statistics

Statistic	Value
Mean	66.54
Std. Dev.	10.88
Minimum	25.70
25th percentile	59.10
Median	66.50
75th percentile	74.00
Maximum	100.00

Table 4: final_grade distribution

Grade	Count	Percentage
C	10,005	50.02%
B	4,222	21.11%
D	4,149	20.74%
F	1,344	6.72%
A	280	1.40%

The 93.28%/6.72% imbalance in pass_fail (a ratio of **13.88:1**) is the single most important structural property driving the metric choices in Section 6 — plain accuracy is inadequate for evaluating any model on this target.

Table 5: `pass_fail` distribution

Class	Count	Percentage
Pass	18,656	93.28%
Fail	1,344	6.72%

3.3 Outlier and Plausibility Checks

An IQR-based scan ($1.5\times$IQR rule) flagged outliers in only a small number of columns, with `final_score` showing the largest count at 68 flagged rows (0.34% of records) — negligible enough that no removal or winsorizing was applied. Plausibility checks (e.g. attendance $\in [0, 100]$, sleep hours $\in [0, 24]$) confirmed no impossible values remained after the cleaning stage described in Section 2.

4 Data Leakage Audit

4.1 Motivation

A core requirement of this project was to formally verify — rather than assume — whether final_grade and pass_fail could be trivially derived from final_score. Using either as a raw predictor for the other would produce artificially perfect results that carry no real-world modeling value.

4.2 Empirical Findings

Contrary to an initial working hypothesis that these labels might be *noisy* derivatives of final_score, the audit revealed they are in fact **exact, deterministic threshold functions** of the score:

Table 6: Deterministic final_score → final_grade mapping

Grade	final_score Range
A	90.0 – 100.0
B	75.0 – 89.9
C	60.0 – 74.9
D	50.0 – 59.9
F	25.7 – 49.9

pass_fail splits cleanly at **final_score = 50**: the maximum score among Fail records was 49.9 and the minimum among Pass records was 50.0, with **zero rows** in any ambiguous overlap zone. A single-threshold decision rule reproduces pass_fail with over 99.8% accuracy.

4.3 Leakage Classification

Every feature was classified into one of three categories:

Table 7: Leakage classification summary

Category	**Definition**
A — Safe	Information realistically known before the prediction point (34 of 38 candidate features).
B — Potential leakage	midterm_score, quiz_avg_score — safe only under the post-midterm prediction-time framing adopted in this project.
C — Direct leakage	final_grade, pass_fail — exact deterministic derivatives of the target; excluded unconditionally.

4.4 Prediction-Time Cutoff

The prediction point is formally defined as **immediately after the midterm exam, before the final exam**. Under this definition, midterm_score and quiz_avg_score are legitimately available and are retained as predictors, while final_score, final_grade, and pass_fail are excluded without exception. This exclusion was enforced in code via an explicit assertion that fails if any leakage column enters the feature set — a directly verifiable, not merely documented, safeguard.

5 Exploratory Data Analysis

5.1 Correlation with the Target

Pearson correlation of every numeric feature against `final_score` identified a sharp divide between a small set of informative features and a large set of near-independent ones.

Table 8: Correlation of key features with `final_score`

Feature	Pearson r
`midterm_score`	0.634
`quiz_avg_score`	0.378
`lecture_attendance_rate`	0.311
`assignment_submission_rate`	0.309
All other numeric features	$\lvert r \rvert < 0.05$

Contrary to common intuition, variables often assumed to be influential — previous GPA, stress level, sleep hours, motivation, exam anxiety, study hours per week — showed negligible linear association with final performance in this dataset.

5.2 Statistical Verification (Welch's t-test)

Mean differences between Pass and Fail groups were tested formally rather than read off a chart:

Table 9: Pass vs. Fail group means for the four strong predictors

Feature	Pass Mean	Fail Mean	t	p
`midterm_score`	51.91	17.90	72.27	≈ 0
`quiz_avg_score`	50.74	27.92	35.72	2.97×10^{-208}
`lecture_attendance_rate`	75.13	65.35	28.42	1.91×10^{-144}
`assignment_submission_rate`	75.03	65.11	28.43	1.83×10^{-144}

For the weak-correlation features, the same test surfaced an important methodological caution: `sleep_hours` ($p = 0.037$) and `motivation_level` ($p = 0.046$) reached nominal statistical significance at $n = 20{,}000$, but their actual mean gaps were practically trivial (6.49 vs. 6.61 hours; 4.99 vs. 4.82 on a 0–10 scale). **At large sample sizes, statistical significance and practical importance are not the same thing** — correlation magnitude (Table 8), not the p-value alone, was used to guide feature-importance conclusions.

5.3 Multicollinearity

A pairwise correlation scan among all numeric features found no pair exceeding $\lvert r \rvert > 0.5$, indicating multicollinearity was not a concern requiring regularization beyond the standard Ridge/L2 penalty already used in the linear baselines.

5.4 Subgroup Exploration

Pass rates were nearly identical across course type (Elective 93.11% vs. Mandatory 93.46%), an early indication — later confirmed formally in Section 9 — that this particular subgroup split does not present a major fairness concern.

6 Feature Engineering

Four engineered features were constructed, each with an explicit rationale, and each was tested against `final_score` rather than assumed to be useful:

Table 10: Engineered features and their tested correlation with `final_score`

Feature	Rationale	Pearson r
`engagement_composite`	Mean of attendance, submission, and lab-participation rates	0.254
`academic_risk_score`	Sum of failed courses and academic warnings	-0.003
`study_effort_per_difficulty`	Study hours relative to course difficulty	0.003
`wellbeing_index`	Sleep hours minus average stress/anxiety	-0.001

Only `engagement_composite` showed meaningful correlation — unsurprising, as it directly aggregates two of the four already-strong predictors identified in Section 5. The remaining three engineered features were retained in the pipeline for interpretability and practical advisory framing, but their near-zero correlation was reported transparently rather than reframed as a positive finding.

7 Modeling Methodology

7.1 Train/Test Split

The data was split 80/20 with **stratification on `pass_fail`**, ensuring the rare Fail class was proportionally represented in both partitions given the 13.88:1 imbalance. The resulting test set contained 4,000 students, including 269 actual Fail cases. The test set remained untouched until final evaluation.

7.2 Preprocessing Pipeline

A `scikit-learn` `ColumnTransformer` + `Pipeline` architecture was used: numeric features were median-imputed and standard-scaled; categorical features were mode-imputed and one-hot encoded. All transformations were fit exclusively on training folds, preventing test-set information from leaking into preprocessing statistics.

7.3 Models Evaluated

Table 11: Models evaluated per task

Target	Model	Role
`pass_fail`	Dummy (majority class)	Naive floor baseline
	Logistic Regression	Interpretable linear baseline (class-weighted)
	Random Forest	Bagged tree ensemble (class-weighted)
	LightGBM	Gradient-boosted trees, hyperparameter-tuned
`final_score`	Dummy (mean)	Naive floor baseline
	Ridge Regression	Interpretable, L2-regularized linear baseline
	Random Forest / LightGBM	Non-linear ensemble alternatives

7.4 Cross-Validation and Hyperparameter Tuning

Model comparison used 5-fold **Stratified K-Fold** cross-validation on the training set, scoring on ROC-AUC, PR-AUC, recall, and F1. The gradient boosting model was further tuned via `RandomizedSearchCV` (15 parameter combinations $\times$ 5 folds), optimizing specifically for **PR-AUC** — a metric more sensitive to rare-class performance than ROC-AUC on an imbalanced target.

8 Results — Primary Task: Pass/Fail Classification

8.1 Final Comparison Table

Table 12: Test-set performance — Pass/Fail classification

Model	ROC-AUC	PR-AUC	Recall	F1	MCC
Logistic Regression	**0.9443**	**0.6125**	**0.881**	**0.448**	**0.461**
Random Forest	0.9311	0.5385	0.093	0.167	0.266
LightGBM (tuned)	0.9384	0.5976	0.859	0.442	0.451

Key finding: the class-weighted Logistic Regression baseline outperformed the hyperparameter-tuned LightGBM model on every single evaluation metric, despite the latter undergoing a 75-fit randomized search. This is consistent with the EDA finding (Section 5) that real signal is concentrated in a small number of roughly linear predictors — precisely the regime in which linear models are competitive with, or superior to, non-linear ensembles. **Logistic Regression is therefore recommended as the final production model** for this task.

Random Forest is clearly the weakest candidate: despite class weighting, its recall on the Fail class (0.093) means it misses the overwhelming majority of at-risk students, making it unsuitable for an early-warning application regardless of its respectable ROC-AUC.

8.2 Detailed Metrics — Final Model (Tuned LightGBM, for comparison)

Table 13: Full classification report — tuned LightGBM, test set

Class	Precision	Recall	F1-score	Support
Pass	0.99	0.85	0.92	3,731
Fail	0.30	0.86	0.44	269
Accuracy		0.854 (4,000 samples)		

Additional metrics: Balanced Accuracy 0.856, Log Loss 0.297, Brier Score 0.096 — indicating reasonably well-calibrated predicted probabilities.

8.3 Confusion Matrix and Cost Asymmetry

On the 4,000-student test set, the tuned model produced:

- **False Negatives: 38** — at-risk students missed entirely.
- **False Positives: 546** — Pass students unnecessarily flagged.

In an early-warning context, False Negatives are the costlier error: a missed at-risk student receives no intervention at all, whereas a false positive merely results in an advisor checking in on a student who turns out to be fine. This asymmetry is why the project prioritizes **recall** over precision throughout.

8.4 Threshold Sensitivity

Because the decision threshold need not default to 0.5, a sweep was performed to quantify the precision/recall trade-off:

Table 14: Precision/recall trade-off across decision thresholds

Threshold	Precision	Recall	F1
0.50	0.297	0.859	0.442
0.40	0.266	0.903	0.412
0.30	0.240	0.941	0.383
0.25	0.224	0.948	0.362
0.20	0.208	0.955	0.341
0.15	0.188	0.967	0.315
0.10	0.166	0.974	0.284

An institution with sufficient advising capacity may reasonably lower the operating threshold toward 0.15–0.20, catching over 95% of at-risk students at the cost of a higher false-alarm rate.

9 Error Analysis

9.1 False Negative Profile

The 38 missed at-risk students were compared against the overall test-set averages on the four strongest predictors:

Table 15: False negative students vs. overall test-set means

Feature	False Negative Mean	Overall Test Mean
`midterm_score`	38.39	48.62
`quiz_avg_score`	40.21	49.41
`lecture_attendance_rate`	66.37	74.45
`assignment_submission_rate`	71.61	74.32

Missed students score below the test-set average on all four predictors but not dramatically so — the predicted `P(Fail)` for these 38 students had a mean of **0.289** (median 0.316), confirming they are **borderline cases sitting near the decision boundary** rather than confidently-wrong predictions. This supports the threshold-lowering strategy discussed in Section 8.4 as a practical mitigation.

9.2 Worst Regression Errors

For the secondary regression task, the largest absolute errors (up to ≈23 points) occurred for students with inconsistent feature profiles — e.g. very low midterm/quiz scores paired with high assignment submission rates, or the reverse — indicating the model struggles most with students whose behavioral signals point in conflicting directions, a reasonable and explainable failure mode rather than a systematic flaw.

10 Results — Secondary Task: Final Score Regression

Table 16: Test-set performance — `final_score` regression

Model	MAE	RMSE	R^2
Dummy (mean)	8.75	—	≈ 0.000
Ridge Regression	**4.358**	**5.535**	**0.7446**
Random Forest	4.532	—	0.7252
LightGBM	4.445	—	0.7353

The pattern mirrors the classification task exactly: the interpretable linear model (Ridge) matched or slightly outperformed both tree-based ensembles, reinforcing that the true relationship between the four strong predictors and `final_score` is close to linear. The model explains approximately 74–75% of the variance in final score — a credible figure given that only a handful of features carry genuine signal.

11 Fairness & Robustness Analysis

Model performance was evaluated separately across subgroups to check for uneven treatment rather than relying on an aggregate metric alone.

Table 17: Subgroup performance — tuned classification model

Subgroup	n	Actual Fail Rate	Recall (Fail)	ROC-AUC
Male	2,043	6.6%	0.881	0.940
Female	1,957	6.8%	0.836	0.937
Part-time job: Yes	1,899	7.3%	0.833	0.930
Part-time job: No	2,101	6.2%	0.885	0.946
Elective course	2,107	6.9%	0.870	0.935
Mandatory course	1,893	6.5%	0.846	0.942

All subgroup recall values fall within a $\sim$5-point band (0.833–0.885) and ROC-AUC within a $\sim$1.6-point band (0.930–0.946) — consistent with normal sampling variation given subgroup sizes of $\sim$1,900–2,100, rather than a systematic fairness issue. The relatively lowest recall was observed for students with a part-time job (0.833), which — while not a severe disparity — is worth ongoing monitoring if the system is deployed operationally.

12 Explainable AI

SHAP (SHapley Additive exPlanations) was used to interpret the tuned LightGBM model at both the global and individual-student level, providing a transparency layer regardless of which model is ultimately deployed.

12.1 Global Interpretation

The SHAP global summary confirmed that the same features driving the correlation and statistical-test findings in Section 5 also dominate the model's actual decision-making, in this order of importance:

1. `midterm_score`
2. `quiz_avg_score`
3. `lecture_attendance_rate`
4. `assignment_submission_rate`
5. `engagement_composite` (engineered feature, secondary explanatory value)

12.2 Local Interpretation

For individual students, SHAP force plots decompose a specific prediction into per-feature contributions — showing, for example, exactly which combination of a low midterm score and poor attendance pushed a particular student's predicted Fail-risk above the decision threshold.

12.3 Important Caveat

Throughout this analysis, SHAP values are interpreted strictly as a description of **what the model relied on to make its prediction** — they are explicitly *not* treated as evidence of a causal driver of student failure. The dataset design does not support causal claims, and this distinction is maintained consistently in every interpretation offered to end users of the system.

13 Deployment: Power BI Dashboard

To make the analysis actionable for non-technical stakeholders (academic advisors and administrators), the cleaned dataset and key derived metrics were published as an interactive **Power BI dashboard** — the "StudiX Dashboard" — shown in Figure 1.

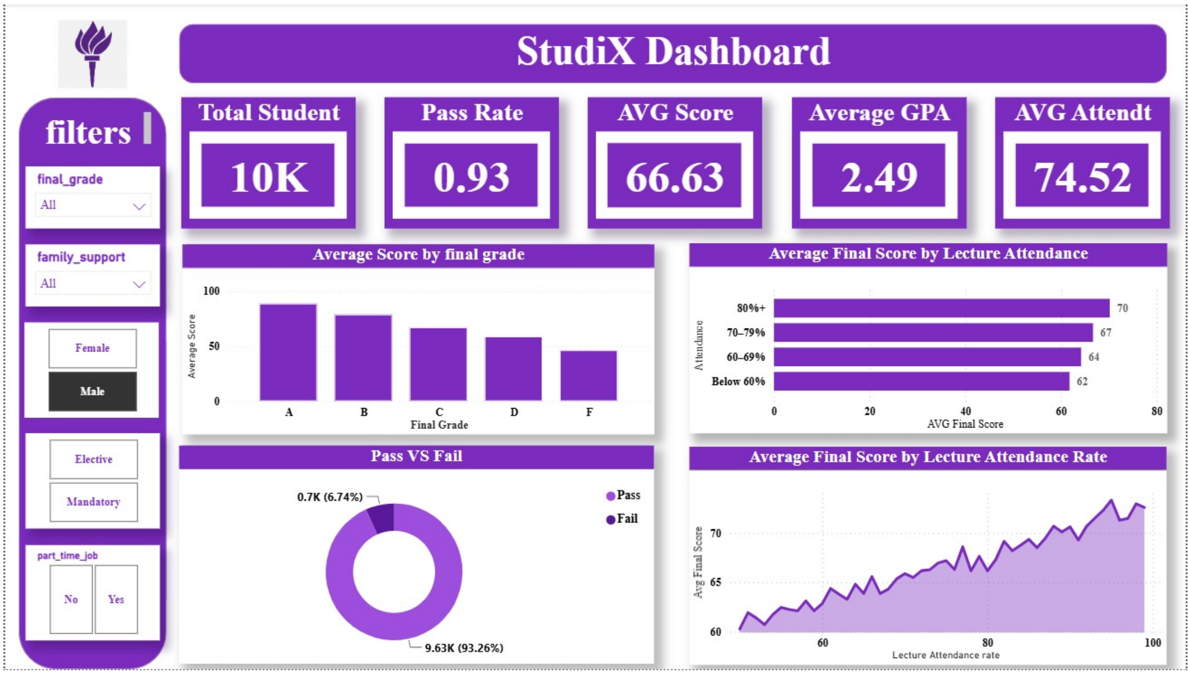

Figure 1: StudiX Power BI dashboard: KPI summary cards, grade/attendance breakdowns, and interactive filters for academic advisors.

13.1 Dashboard Components

- **KPI summary cards:** Total Students (10K sample view), Pass Rate (0.93), Average Score (66.63), Average GPA (2.49), and Average Attendance (74.52) — giving advisors an at-a-glance cohort health check consistent with the dataset-wide statistics reported in Section 3.
- **Average Score by Final Grade:** a bar chart confirming the expected monotonic relationship between grade band and average score, visually reinforcing the deterministic mapping formally verified in Section 4.
- **Pass vs. Fail donut chart:** directly visualizes the 93.26%/6.74% class split (matching the 93.28%/6.72% audited split in Section 3 to within dashboard-sample rounding), keeping the core imbalance visible to every dashboard user.
- **Average Final Score by Lecture Attendance (bar and trend views):** makes the attendance–performance relationship identified statistically in Section 5 immediately visible to a non-technical audience — average score rises steadily from 62 (below 60% attendance) to 70+ (80%+ attendance).
- **Interactive filters:** final grade, family support level, gender, course type, and part-time job status — allowing advisors to slice the same subgroup views formally analyzed for fairness in Section 11.

13.2 Role in the Overall System

The dashboard closes the loop between the modeling work in Sections 7–10 and its intended end users: rather than requiring an advisor to interpret a notebook or a raw probability score, the dashboard translates the same underlying signals (attendance, scores, pass/fail split) into a filterable, visual format suited to day-to-day advising decisions. In a full production deployment, the model's per-student risk scores (Section 8) would be integrated as an additional dashboard field, allowing advisors to filter directly by predicted risk category alongside the existing descriptive filters.

14 Conclusion

14.1 Summary of Findings

1. The dataset was clean at the structural level (no missing values, no duplicates) but required careful, per-column governance decisions around imputation and target-variable handling.
2. `final_grade` and `pass_fail` are exact deterministic derivatives of `final_score` and were rigorously excluded as predictors.
3. Real predictive signal is concentrated in four features — midterm score, quiz average, lecture attendance, and assignment submission rate — while most demographic and psychological variables showed negligible relationship with outcomes in this dataset.
4. A simple, interpretable Logistic Regression model outperformed a tuned LightGBM gradient boosting model on every test-set metric for the primary classification task, and Ridge Regression similarly led the secondary regression task — both consistent with the near-linear structure revealed by the EDA.
5. The final recommended model achieves 88.1% recall on at-risk students, with a tunable decision threshold allowing the precision/recall trade-off to be adjusted to institutional capacity.
6. Subgroup fairness analysis found no severe disparities across gender, part-time job status, or course type.

14.2 Limitations

The dataset's uniformly-distributed "noise" features and perfectly deterministic target-derivation are consistent with synthetic data generation; real institutional data may exhibit noisier, more genuinely non-linear relationships. No student or course identifier exists in the dataset, preventing longitudinal or repeated-measures analysis. The model cannot support causal claims about *why* students underperform — only which observable factors correlate with risk.

14.3 Ethical Considerations

This system is explicitly designed as a **decision-support tool**, not an automated judgment of a student's future. All flagged predictions are intended for human review by an academic advisor. Labeling a student as "at risk" carries real reputational and psychological stakes and should be communicated supportively, with explainability (Section 12) used to keep the reasoning behind any flag transparent and contestable.

14.4 Future Work

- Collect finer-grained, week-by-week behavioral data rather than semester aggregates, to enable earlier and more precise risk detection.
- Introduce student/course identifiers to support longitudinal, repeated-measures validation.
- Integrate live model scoring directly into the Power BI dashboard as a risk-category filter.
- Pursue a causal-inference study design if the project's goal evolves from prediction toward evaluating specific intervention strategies.
- Validate whether the linear model's advantage persists on real (non-synthetic) institutional data.